Name(s): ____________________

DATA 101 Assignment 5: Python

Work in teams of 2

Instructions. Create a new Jupyter Notebook. Remember to use the cell types Markdown for text explanations/titles and Code for Python code. Show all your text explanations and code. Once you complete the assignment save your file as `python_is_fun.ipynb`. Print your jupyter notebook file and bring your assignment with you to class on Monday. To export your notebook, you can click on `File → Save and export notebook as → HTML (.html)` (don't forget to allow for pop-ups) then print the `.html` file using any browser. (you can double click on an `.html` file.)

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1. Convert your first cell into Markdown type and do the following:

(a) Write a bold title: `Here begins the assignment`.

(b) Add a short italicized motto for yourself (e.g., could be short like *Code* > *Sleep*, or something longer like a quote).

(c) Make a bulleted list of your 3 hobbies.

(d) Write and display 3 of your favorite emojis. (e.g., 🐍, 🎵, ⚽)

2. Suppose you are budgeting for snacks. A burger costs 6.75, fries 2.50, and a soda 1.75.

(a) Create variables for each item, and compute the total cost for 3 burgers, 2 fries, and 4 sodas.

(b) Let x, y, and z be the number of burgers, fries, and sodas you want to order. Write code that multiplies each by its price and computes the new total.

(c) Write an if–else that prints `"too expensive"` if the total from (b) is greater than \$25, else print `"affordable"`.

3. (a) Write a for loop that prints:

```
Python is fun! (1)
Python is fun! (2)
Python is fun! (3)
Python is fun! (4)
Python is fun! (5)
```

(b) Write a for loop that prints the squares of numbers 1 through 10 in the form:

```
1 squared is 1
2 squared is 4
...
10 squared is 100
```

4. (a) Write a function `cheer(name)` that takes someone's name and returns a string like `"Go Eren!!!"`. Test it with three different names.

(b) Write a function `echo(word, times)` that repeats a word the given number of times with spaces in between. Example: `echo("hi", 3)` → `"hi hi hi"`.

5. (a) Import NumPy. Simulate rolling a six-sided die 100 times. Show the first 10 rolls and compute the mean roll. (*Hint: Use np.random.randint()*). Read more here.

(b) Simulate heights of 200 people with mean = 170 cm and SD = 10. Round to 1 decimal and print the tallest height. (*Hint: Use the numpy function we used in class.*)

6. Read in the following data from Mario Kart to your python session: `https://github.com/rfordatascience/tidytuesday/raw/refs/heads/main/data/2021/2021-05-25/records.csv` Name your dataframe object as `df`.

(a) Show the first 7 rows.

(b) Show how many rows and columns the dataset has.

(c) Find the mean `record_duration`.

(d) Repeat (c) but for the Rainbow Road track only.

(e) Repeat (c) for `type = Single Lap and Three Lap`.

7. (a) Make a histogram of `record_duration`. Choose a color you like and make sure to label your axes.

(b) Make a scatterplot of x=`time` and y=`record_duration`. Choose a color your like for the dots and make sure to label your axes.